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Review

Towards Predictive Maintenance in the Maritime Industry: A Component-Based Overview

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Abstract: The maritime industry has a significant influence on the global economy, underscoring the need for operational availability and safety through effective maintenance practices. Predictive maintenance emerges as a promising solution compared to conventional maintenance schemes currently employed by the industry, offering proactive failure predictions, reduced downtime events, and extended machinery lifespan. This paper addresses a critical gap in the existing literature by providing a comprehensive overview of the main data-driven PdM systems. Specifically, the review explores common issues found in vessel components (i.e., propulsion, auxiliary, electric, hull), examining how different state-of-the-art PdM architectures, ranging from basic machine learning models to advanced deep learning techniques aim to address them. Additionally, the concepts of centralized machine learning, federated, and transfer learning are also discussed, demonstrating their potential to enhance PdM systems as well as their limitations. Finally, the current challenges hindering adoption are discussed, together with the future directions to advance implementation in the field.

Keywords: predictive maintenance; machine learning; maritime industry; Shipping 4.0

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1. Introduction

The maritime industry has a significant influence on the global economy, facilitating the transportation of over 80% of trade goods. According to the 2024 UN Trade and Development report, more than 12 million cargo shipments were transported only last year, with projections showcasing a steady annual increase of 2.4% by 2029. This underscores the critical need for operational availability and safety of the vessels, which is highly correlated with maintenance operations [1].

Despite the industry's adoption, traditional maintenance strategies, such as Reactive Maintenance (RM) and Preventive Maintenance (PM), have shown limitations in preventing accidents, often contributing to significant expenses. For instance, only in 2023, half of reported maritime accidents were attributed to mechanical failures [2,3]. Towards addressing these challenges, research efforts have focused on developing Predictive Maintenance strategies (PdM). Unlike RM and PM, PdM leverages Machine Learning (ML) models to proactively forecast potential equipment degradation or failures based on vessel-related operational data-driven insights.

The concept of ML-driven PdM is integral to the fourth generation of shipping, known as Shipping 4.0. This paradigm envisions the integration of novel technological advancements such as the Internet of Ships (IoS) and Artificial Intelligence (AI) to transform traditional ships into "smart ships". Within this paradigm, these advancements will enable automated fleet management, reduce downtime events, and meet environmental regulations, contributing to a more efficient and sustainable sector [4,5]. However, despite these advancements, real-world deployment challenges remain, including model scalability, communication efficiency, and data privacy and security, especially in conventional Centralized Machine Learning (CML) approaches. In detail, CML requires the uploading of vast amounts of fleet data to central servers, increasing the cyberattack surface and operational risks, while also having to adapt to the different vessel types and sizes. To mitigate these issues, decentralized learning paradigms such as Federated Learning (FL) have been recently introduced, ensuring secure data gravity, enabling ships to act as edge nodes. FL enables the localized training of ML models onboard ships, acting as clients. The FL server collects and aggregates model parameters, ensuring privacy, training models without requiring the transmission of raw data to the server through iterative rounds, and providing a global model tailored to vessel-specific conditions [6–8].

Recent studies have examined aspects of fault prediction in the maritime industry. For instance, Ali et al. focused on corrosion detection systems for hull structures [9], Zhang et al. explored RUL prediction models [10], while Liang et al. focused on the regulatory role of maritime classification societies [11]. Although these studies offer significant information on the different aspects of PdM, they lack a comprehensive analysis of PdM approaches by the different vessel systems. To the best of our knowledge, there is no existing work that provides a comprehensive overview of PdM approaches spanning all major vessel systems.

Taking into account the significant potential of integrating PdM strategies in the maritime domain and the literature gap, this work aims to provide an overview of the latest data-driven PdM techniques applied across various target vessel systems (i.e., propulsion, electrical, auxiliary, hull), identifying trends while also mapping the relevant open issues and potential future directions towards stimulating further research investigations.

Figure 1 showcases the structure of this paper. Section 2 introduces in detail the different maritime maintenance strategies, while Section 3 categorizes and presents state-of-the-art maritime PdM approaches. Then, Section 4 discusses open challenges and their respective future directions, while finally Section 5 concludes our paper.

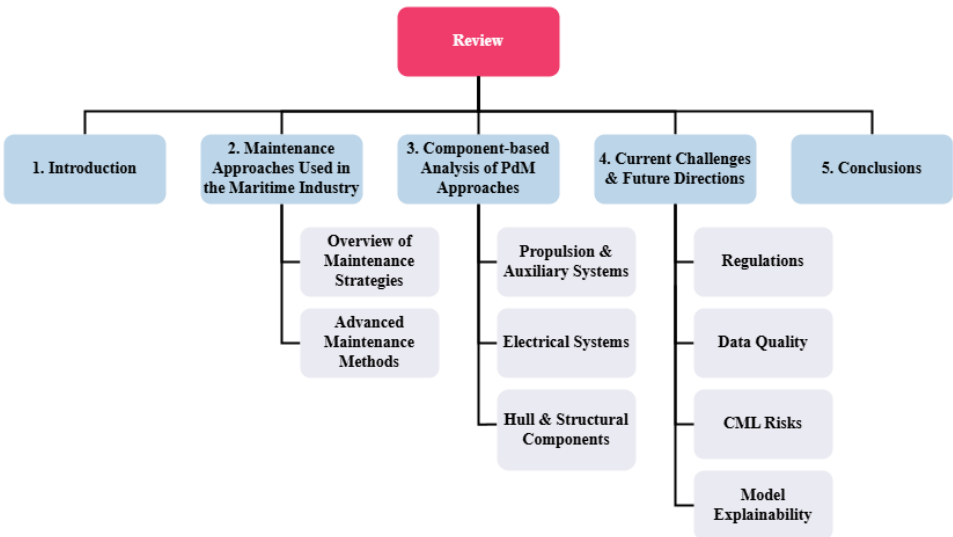

Figure 1. Review structure.

2. Maintenance Approaches Used in the Maritime Industry

2.1. Overview of Maintenance Strategies

Maintenance operations in the maritime domain have traditionally relied on RM. RM enables repairs to be performed directly after a failure has occurred as a response, taking place either onboard or during shipyard visits. While RM addresses immediate issues, it has been recognized as costly and inefficient (i.e., mostly due to unexpected downtime events), focusing on corrective actions rather than preventing ones, raising safety concerns. Additionally, due to the high total operational costs (OPEX) (i.e., around 40%) associated with the maintenance operations of a vessel, maritime organizations have sought to explore alternative optimized strategies [12,13].

Furthermore, according to the International Convention for the Safety of Life at Sea (SOLAS), maritime organizations are obliged to conduct a series of inspections, including an initial survey, followed by annual, intermediate, and renewal surveys conducted every five years within a vessel's operational lifecycle [14,15]. Towards addressing this, PM strategies have been introduced, targeting different vessel components. In detail, a PM strategy involves the creation of scheduled maintenance inspections based on either industry-standard intervals or manufacturer-provided degradation projections. Although PM reduces unexpected breakdowns and downtime events compared to RM, it often leads to substantial OPEX associated with unnecessary repairs and overscheduled maintenance periods [16].

In contrast to RM and PM, PdM aims to increase equipment lifespan, reducing breakdowns, while optimizing inspections to minimize downtime events and costs using data-driven methods such as ML. Specifically, PdM leverages data coming from maritime vessels (e.g., vibrations, temperature, etc.), generating predictions of expected equipment failures. Studies indicate that PdM has the potential to minimize maintenance OPEX by as much as 45% compared to conventional maintenance strategies, while also reducing safety risks and alleviating crew workload [17,18].

2.2. Advanced Maintenance Methods: Model-Based and Machine Learning-Based

This section describes the different methods utilized towards PdM in the maritime industry. Toward this, two main categories of methods are distinguished, physics-based models and data-driven models, encompassing ML and Deep Learning (DL) models.

2.2.1. Physics-Based Methods for Predictive Maintenance

PdM aims to optimizing the time-consuming maintenance intervals and vetting inspections by predicting potential operational failures. Physics-based methods rely on mathematical models specifically designed to estimate the degradation state of a component based on the observed system values. While they can be advantageous due to their simplicity and minimal computation demands, they can, however, make it challenging to capture all the complex physics-related intricacies of a system. In addition, potential changes in the operational conditions of the systems or the effects of factors not included in the model can affect the accuracy of forecasts. Notable model-driven methods include Kalman Filters (KFs) and Weibull analysis [11,19–21].

2.2.2. Data-Driven-Based Methods for Predictive Maintenance

Currently, the majority of operational vessels have undergone a digital transformation aligning with Shipping 4.0 principles [22]. These vessels are often referred to as Cyber-Enabled Ships (CESs) functioning within a comprehensive ecosystem that includes (i) the vessel, (ii) a Shore-based Control Center (SCC), responsible for managing the CES, and (iii) the communication protocols used between the CES and the SCC. Furthermore, the CES

comprises both Information Technology (IT) and Operational Technology (OT) systems, essential for the vessel's operations [23]. Based on these IT and OT systems, vast operational data are generated and collected everyday by onboard sensors. AI has made it possible for maritime organizations to harness the potential of this data effectively, enabling them to meet diverse service requirements (e.g., PdM, fuel consumption forecasting, etc.), enabling optimal decision making [24,25]. Unlike traditional model-based methods, ML-driven methods are considered more suitable for state-of-the-art PdM models, offering the ability to (i) learn patterns from complex and large datasets; (ii) adapt to machines without the need for designing detailed physical models; (iii) make accurate prediction in real-time; and (iv) be robust to noisy and incomplete data [26–28].

ML is a subset of AI focusing on the development of intelligent models capable of learning from different patterns and insights processed from input data to make decisions [29]. ML is divided into three main categories, namely supervised learning, unsupervised learning, and Reinforcement Learning (RL), where:

- Supervised Learning relies on labeled datasets comprising both the input (X) and its corresponding output label (Y) to learn a mapping function ($f : X \rightarrow Y$). Example supervised tasks in the maritime domain include classification and regression used, respectively, for the detection of machinery failure and for the proactive prediction of potential failures.
- Unsupervised Learning deals with the discovery of patterns within data lacking the labeled outputs. Widely used techniques within unsupervised learning include clustering and dimensionality reduction. Clustering enables the grouping of similar data points, while dimensionality reduction enables the reduction of data dimensions, while preserving their structure. Within the maritime domain, unsupervised learning algorithms can be used to identify patterns concerning fuel usage and abnormal mechanical behavior or to simplify complex maritime data for analysis and visualization.
- RL focuses on learning a policy on a specific environment, taking into consideration the costs and cumulative rewards with each action. Example applications of RL within the maritime domain include autonomous vessel navigation and PdM inspections scheduling optimization.

Furthermore, current ML applications utilize Maritime Communication Networks (MCNs) to enable wirelesses data transmissions provided by SCC (e.g., company headquarters) and satellite systems. However, these topologies face challenges like communications delays, spectrum availability, and intermittent connectivity issues [30]. Additionally, CML applications introduce significant security and scalability concerns. For instance, as CML models require the vessel-to-shore communication to transmit operational data to a server, there are risks concerning the likelihood of potential unauthorized access or misuse of ship data used for training the ML models. Moreover, the incorporation of multiple vessels that generate large amounts of data can result in scalability challenges [31,32].

Toward this end, FL has been introduced, enabling vessels to act as edge nodes (i.e., FL clients) and train models onboard aiming to collaboratively solve a learning task, transmitting only model parameters and their data. This process is orchestrated by an FL server, which updates and aggregates the global model back to the clients based on a predefined number of rounds. FL's decentralized training approach enhances data privacy and security, reducing the cyber-attack surface compared to CML, while supporting model scalability by including more client nodes (i.e., ships) in the training process, making it highly suitable for maritime applications like PdM [5,33].

However, in real-world applications, the availability of labeled data is often limited, with their collection being considered costly, time-intensive, or impractical, particularly in situations dealing with legacy systems. Transfer Learning (TL) addresses these challenges,

by enhancing the performance of ML models through knowledge transfer from related source domains, reducing the need for extensive operational data [34]. In the context of maritime PdM, TL provides a solution for scaling models to accommodate varying component manufacturers and technologies. For example, knowledge gained from a hybrid DE of a bulk carrier can be transferred to a single fuel DE of a passenger ship. Figure 2 illustrates the differences among CML, FL, and TL training.

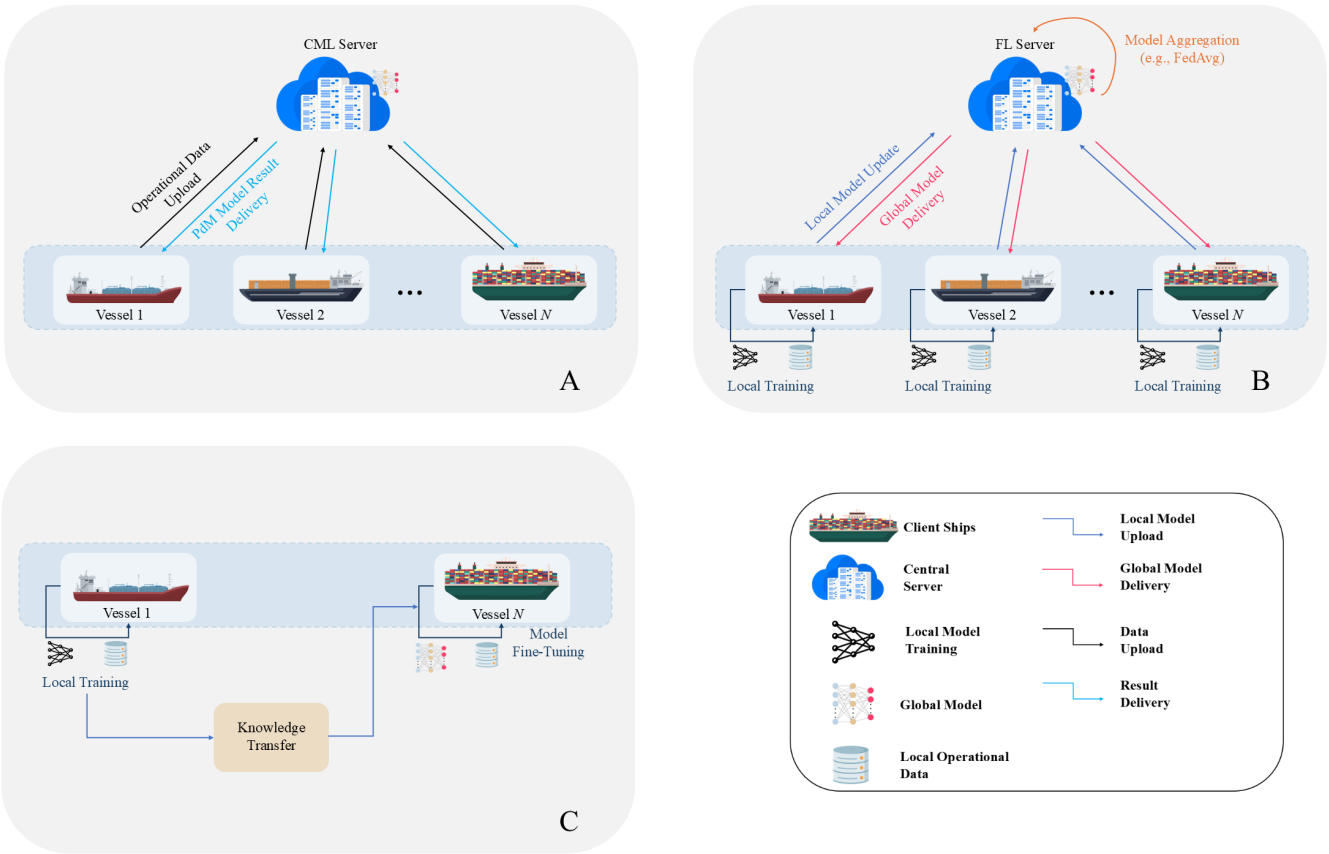

Figure 2. Comparison of ML training schemes: (**A**) CML, (**B**) FL, and (**C**) TL.

2.2.3. Techniques in Predictive Maintenance

PdM represents a transformative approach to asset management in the maritime industry. Leveraging data-driven insights can diagnose the state of the equipment and its issues, as well as prognose potential future failures, toward prolonging the equipment lifespan [35,36]. In this context, different techniques can be found in the PdM literature, each bringing unique strengths and addressing different challenges and needs, including the following:

- Remaining Useful Life (RUL) is defined as the remaining time for an asset to be able to perform functionally before reaching its End Of useful Life (EOL) [37]. Suppose t_e is the predicted EOL time, while t is the current time, RUL can be expressed as:

$$RUL(t) = t_e - t \tag{1}$$

- Time To Failure (TTF) is defined as the remaining time before an asset reaches failure. Unlike RUL, TTF focuses on calculating the absolute time of failure. The difference

between TTF and RUL is illustrated in Figure 3 [38]. Suppose t_f is the time at which failure occurs, while t is the current time, TTF can be expressed as:

$$TTF(t) = t_f - t \tag{2}$$

- Time To Repair (TTR) is defined as the time required to repair a maritime asset to its normal operating condition [39]. Suppose t_r is the time when the asset is fully restored, while t_f is the failure time, and TTR can be expressed as:

$$TTR(t) = t_r - t_f \tag{3}$$

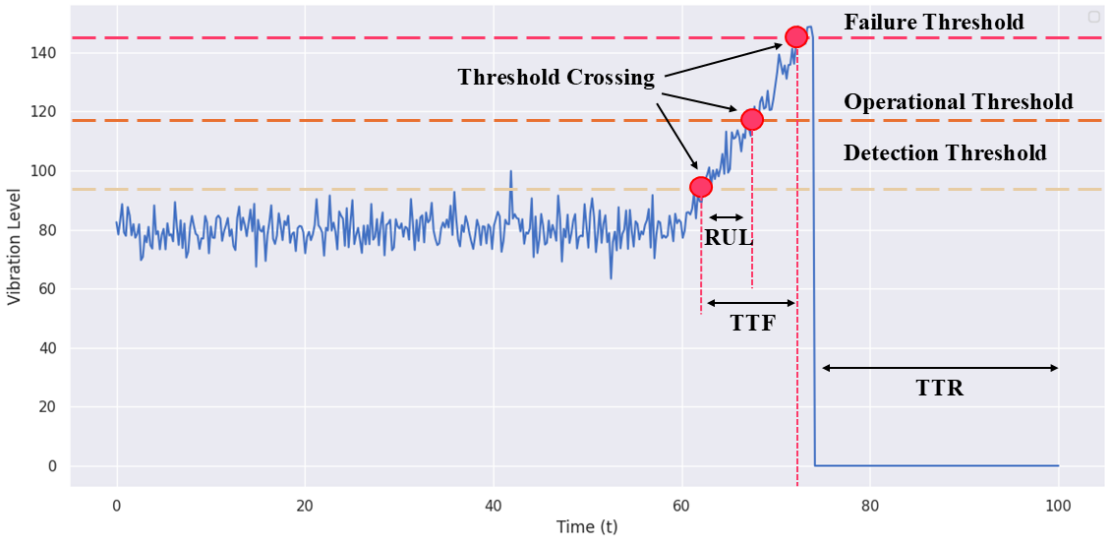

Figure 3. Distinction between RUL, TTF, and TTR.

3. Component-Based Analysis of PdM Approaches

This section presents an in-depth analysis of PdM applications from a component-based perspective, focusing on key systems within maritime vessels.

3.1. Propulsion and Auxiliary Systems

Power generation is integral to a vessel's performance, efficiency, and environmental footprint, making its maintenance of critical importance. Since the 20th century, Internal Combustion Engines (ICEs) and especially marine Diesel Engines (DEs) have been the primary technologies for power generation due to their high operational and economic efficiency. However, as DEs are responsible for high NOx emissions, maritime companies have explored alternative systems to align with the GHG reduction strategy enforced by the International Maritime Organization (IMO) [40–42]. Currently, the industry employs both conventional and hybrid propulsion systems, including the following:

- Single-fuel systems use conventional fuels, such as Marine Diesel Oil (MDO) and Heavy Fuel Oil (HFO), or alternative fuels, like Liquefied Natural Gas (LNG).
- Dual-fuel systems enable flexibility, running on a combination of fuels (i.e., LNG/diesel, methanol/diesel, etc.).
- Hybrid electric (HE) systems combine ICE and batteries to further improve efficiency and reduce emissions. HE systems have been increasingly adopted due to the national and regional regulations introduced in the last few years (e.g., the European Fit for 55 Package [43] and IMO's Sulfur Cap [44]). In detail, HE systems are broadly classified into three main architectures based on the energy administration, namely serial, parallel, and hybrid. Under an HE topology, all vessel subsystems, from the

power generators to the bridge navigational equipment, are connected to the main electrical grid using different converters [40].

Furthermore, auxiliary systems are essential components supporting the primary functions of a vessel, complementing the main propulsion and navigation systems. The main categories of auxiliary systems include (i) fluid systems (e.g., lubrication, fuel, ballast); (ii) Heating, Ventilation, and Air Conditioning (HVAC) systems; (iii) safety systems (e.g., sprinklers, lifeboats, etc.); (iv) waste management systems; (v) navigation and communication support; and (vi) engine room systems (e.g., cooling, air compressors) [45,46].

Propulsion and auxiliary system failures can have a significant impact on safety, environmental sustainability, and economic performance. In general, the factors responsible for failures in propulsion systems can be classified into two main categories, namely unpredictable and predictable factors. Unpredictable factors include severe environmental conditions, operational errors, or even corrosion of key components. Predictable factors include stress factors that can originate from the design or manufacturing stages of components (e.g., material defects, etc.) or due to equipment fatigue over time leading to gradual degradation [47]. Common failures found in the propulsion system are associated with the (i) fuel system (e.g., filter blockages, injection fails, etc.); (ii) lubrication of the system with insufficient oil; (iii) cooling system; and (iv) intake and exhaust system of the engine (e.g., blockages in valves, air filters, etc.) [48–54]. The key factors contributing to propulsion-related failures are showcased in Figure 4, while the main HE architectures are showcased in Figure 5.

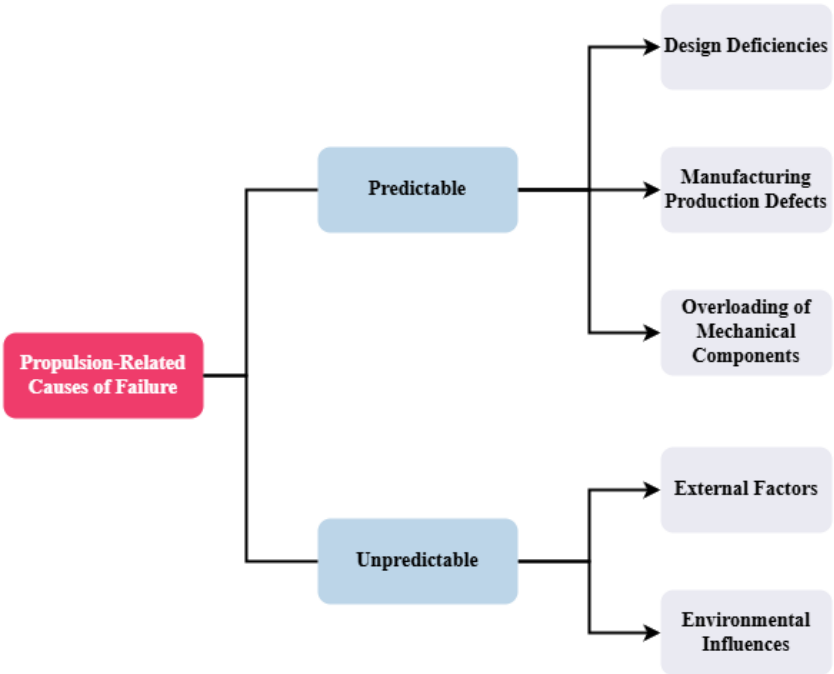

Figure 4. Key factors contributing to propulsion system failures.

Toward assessing the condition of maritime propulsion systems, various ML-driven PdM algorithms can be found in the literature. Linear Regression (LR) models have been used, aiming at estimating data points via a hyperplane. For instance, Cheliotis et al. monitored the state of a 5-cylinder two-stroke engine of a Handymax Bulk carrier by predicting its Exhaust Gas Temperature (EGT) using a polynomial ridge regression model and detecting faults using Exponentially Weighted Moving Average (EWMA) control charts [55]. Similarly, Kang et al. monitored the state of a dual-fuel (diesel–natural gas) engine using Principal Component Analysis (PCA), to minimize data dimensionality and a Least Abso-

lute Residual (LAR) regression model to predict engine load [56]. However, while linear models are relatively easy to implement and can require less data compared to complex DL configurations, they lack the ability to capture complex, non-linear relationships typically found in machinery data. Furthermore, these models are sensitive to outliers, which can result in skewed predictions, reducing their overall reliability in real-world scenarios [57–59].

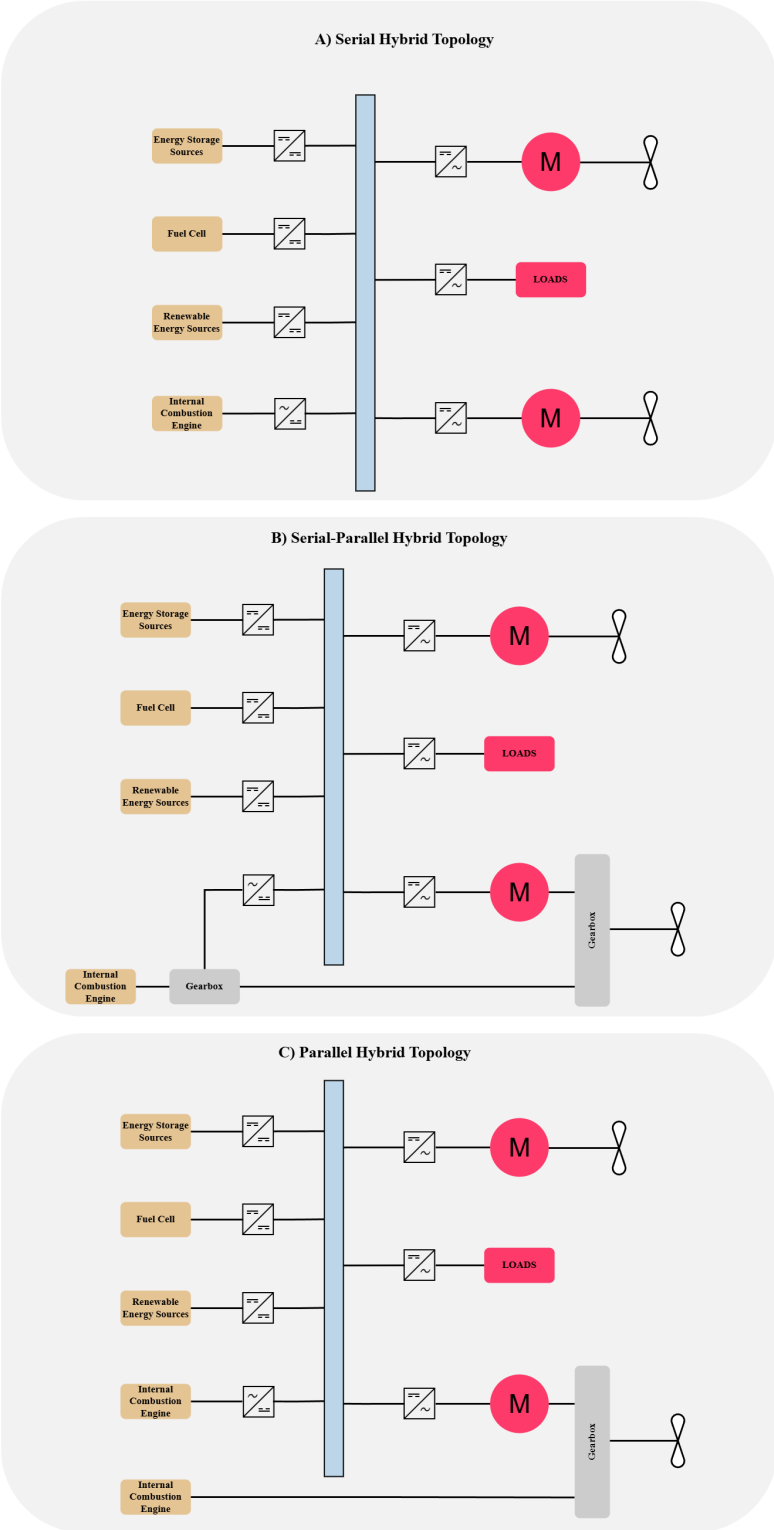

Figure 5. Hybrid propulsion architectural configurations: (**A**) serial hybrid system, (**B**) serial–parallel hybrid system, and (**C**) parallel hybrid system [40].

Furthermore, kernel methods, like Support Vector Machine (SVM), can also be found in the literature. SVM is a supervised learning method aiming to find a hyperplane that optimally separates two classes. Similarly, Support Vector Regression (SVR) extends SVM for regression tasks [60]. Coraddu et al. explored the usage of SVR to forecast the degradation of a Gas Turbine (GT) on a simulated Combined Diesel–Electric and Gas (CODLAG) propulsion system [61,62]. Additionally, Vorkapic et al. explored polynomial and Pearson VII kernel functions on an SVR model to monitor the performance of a two-stroke DE of a Liquefied Petroleum carrier [63]. However, despite their advantages, kernel methods like SVR can also lead to prolonged training and inference times, attributed to their high computational complexity. Additionally, scalability issues may also arise as the dataset size increases, while also being sensitive to the kernel functions, with improper selection affecting their accuracy and generalizability [64,65].

Artificial Neural Networks (ANNs) are inspired by the structure and functionality of biological neural nets. They are composed of three different layers: (i) an input layer, processing the data fed into the network; (ii) hidden layers, performing complex computations; and (iii) an output layer, producing the final result [60]. Different ANN architectures have been utilized for marine PdM scenarios. For instance, Lorencin et al. explored the application of a simple Feed-Forward Network (FNN) Multilayer Perceptron Network, assessing the decay state of a CODLAG propulsion system. The authors conducted extended simulations to determine the optimal model parameters (i.e., activation functions, solvers, network structure, etc.) to improve predictive performance. Results indicated that the use of a Broyden–Fletcher–Goldfarb–Shanno solver and tanh activation function resulted in enhanced performance [66]. Furthermore, Guerrero et al. studied anomalies in a two-stroke low-speed DE based on different loads. Unlike previous works, the authors employed harmonic and sensitivity analysis to link cylinder combustion dynamics to the Fourier coefficients as a first step, to then train an FNN to estimate the mean DE power [67]. ANNs can be suitable for different PdM applications, enabling the modeling of complex data relationships, as well as the processing of different data types (e.g., time-series, images, etc.). However, their drawbacks should also be taken account of, including the requirement for large amounts of labeled historical data to train them, as well as the risk of overfitting without proper regularization [68,69].

Although Recurrent Neural Networks (RNNs) have been widely utilized to deal with sequential data, they lack the ability to learn input data with large time steps. Toward this end, Long Short Term Memory (LSTM) networks have been introduced, effectively handling the issue of long-term dependencies by incorporating gating mechanisms into the RNN structure, making them well-suited for PdM applications [70,70]. Han et al. successfully employed an LSTM network to assess the state of DE by estimating its RUL. The authors evaluated the model in near real-world scenarios against other DL-based alternatives, by not only employing different load profiles but also augmenting the input data using white Gaussian noise [71]. Additionally, Gribbestad et al. explored the application of Transfer Learning (TL) to address data scarcity issues, focusing on predicting the RUL of air compressors. The work utilized an LSTM model fine-tuned with Particle Swarm Optimization (PSO) on a publicly available compressor dataset. Subsequently, components of the pre-trained model were transferred and adapted to a marine air compressor dataset. Experimental results demonstrated that this TL approach significantly outperformed baseline models, highlighting the efficacy of the approach in scenarios with limited run-to-failure data [72]. While LSTM networks can be excellent for PdM applications involving sequential sensor data, capturing their long-term dependencies without performance loss, certain drawbacks should also be taken into account. In detail, LSTMs (i) are resource-intensive (i.e., their high complexity can lead to slower training times), (ii) as a configuration of

ANNs they are also prone to overfitting, especially without sufficient regularization, and (iii) performance can be significantly affected by poorly tuned hyperparameters [70,73].

Convolutional Neural Networks (CNNs) are capable of identifying and extracting features through convolutional operations, primarily used for computer vision tasks [60,74,75]. In the domain of PdM, CNNs have been utilized as part of hybrid architectures to effectively capture temporal dependencies. For instance, Ji et al. developed a hybrid network combining a CNN, a Bidirectional LSTM (BiLSTM) and an attention layer, to assess the state of dual-fuel DE of an LNG carrier based on EGT values. The proposed model leveraged the strengths of (i) CNN to effectively extract features from the input data, (ii) BiLSTM to capture temporal dependencies, and (iii) the attention mechanism to enable the dynamic weight allocation to the input features. The network was evaluated against other baseline and hybrid networks, outperforming them in terms of predictive accuracy [76]. Similarly, Li et al. explored the decay of fuel injections based on data coming from a dual-fuel 9-cylinder 4-stroke medium speed engine, introducing a CNN–Gated Recurrent Unit (GRU)–Convolutional Block Attention Module network to predict the ME RUL [77], while Liu et al., developed a Deep CNN– Bidirectional–GRU model to predict EGT trends of a dual-fuel hybrid electric propulsion system [78]. Research works have also investigated minimizing computation demands for onboard model training. For instance, Kalafatelis et al. evaluated different optimization techniques on a CNN+AE model to predict the RUL of an HFO purifier of a bulk carrier. The study explored techniques, such as early stopping and model pruning, including constant, polynomial decay, and L1 and L2 regularization. Among these, the use of early stopping and L2 regularization demonstrated the most effective performance, balancing predictive accuracy, model complexity, and training time [79]. CNNs can be useful for both time-series (e.g., 1D CNN, hybrid configurations) and image-based PdM tasks. Specifically, CNNs offer the ability to automatically identify spatial or temporal patterns from input data. However, despite their advantages, they require (i) large data sizes to generalize effectively, (ii) high computation resources for training and inference, making edge deployment difficult in FL scenarios, and (iii) careful configuration of their hyperparameters, due to their architectural complexity, especially in hybrid model cases [80–82].

3.2. Electrical Systems

Current research efforts on ship electrification focus on (i) optimizing power systems, (ii) their management, and (iii) electric propulsion systems [83–85]. Central to this are Energy Storage Systems (ESSs), with batteries being the most common type for hybrid ships [40].

Battery-enabled hybrid power system configurations can already be found for various vessel types used for different purposes (e.g., cruise, fishing, and cargo transportation, etc.) to enhance the operating behavior of the main power generators [86,87]. Common power system configuration in hybrid vessels include the traditional DE ICE and lithium-ion cells, such as Lithium–Nickel–Manganese–Cobalt–Oxide (LNMC), Lithium–Iron–Phosphate (LFP) batteries, and Lithium–Titanium–Oxide (LTO) cells, each with its one set of benefits and limitations [88], as follows:

- ICE remains the most common power technology employed today in the maritime industry. Although battery-driven systems offer environmental advantages, ICEs utilizing fuels like HFO or sulfur fuel oil provide higher volumetric energy density. Additionally, ICEs operate on a PM schedule and typically do not need to be replaced during a ship's service life.
- LNMC cells are considered the most common battery type used in maritime vessels, characterized by their high energy capacity.

- Despite having a lower energy capacity at the cell level, LFP cells leverage prismatic form factors, enabling them to have tight packaging, improving energy density at the system level.
- LTO cells are preferred in high-powered hybrid configurations requiring durability under repetitive use, based on their high number of cycles [88].

A summary of the different power system configurations using different energy sources for electric motors is shown in Figure 6.

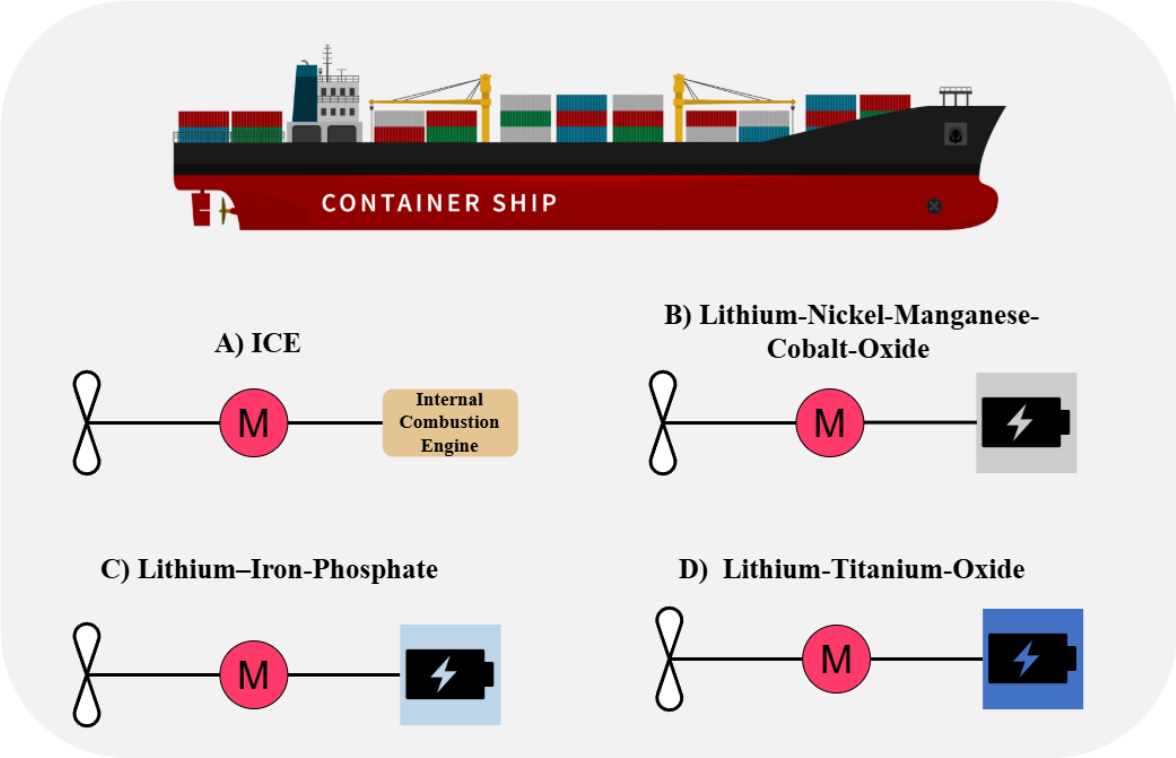

Figure 6. Comparison of traditional and electric power system configurations. (**A**) Internal Combustion Engine (ICE), (**B**) Lithium–Nickel–Manganese–Cobalt–Oxide (NMC) batteries, (**C**) Lithium–Iron–Phosphate (LFP) batteries, (**D**) Lithium–Titanium–Oxide (LTO) batteries [88].

Common issues found in ship electrical systems include but are not limited to (i) cable and wiring damage, (ii) battery failures, (iii) generator, alternator, and transformer malfunctions, and (iv) lighting system failures. Toward addressing these and optimizing performance, several works have explored integrating PdM systems.

For instance, Tang et al. explored degradation states of lithium-ion cells used in hybrid vessels. In detail, their work introduced a Bayesian approach to SVM, named Relevance Vector Machine (RVM), predicting the RUL of the cell capacity [89]. While RVMs share similarities with SVRs in their ability to model non-linear patterns, they also exhibit limitations that hinder their adoption. Specifically, RVMs can be resource-intensive, especially when applied to large datasets, and depend greatly on data preprocessing, as improper scaling can lead to convergence issues [90,91].

Furthermore, Liu et al. developed a hybrid framework to assess the state of marine batteries. The framework integrated the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm to diagnose faults and a transformer to predict voltage anomalies. Transformers are neural networks capable of processing multi-level representations in sequential data, based on self-attention mechanisms [92,93]. The framework was

validated on a large passenger electric ship, showcasing the ability to overcome the delays found in the traditional battery management system [94].

Transformers offer significant advantages for PdM applications, enabling the processing of long data sequences without the memory limitations found in other networks (e.g., RNNs). However, their limitations should also be considered for PdM development, including the following: (i) significant computational and memory resources, (ii) extensive data to generalize effectively and avoid overfitting, and (iii) hyperparameter tuning, which can be challenging and time-consuming [95,96].

Unlike the aforementioned CML approaches, Karandikar et al. proposed an FL-based scheme integrating blockchain technology to further enhance security and privacy. In detail, the framework introduces smart contracts, replacing the traditional FL server, storing and managing the different model parameters on the blockchain. The authors successfully evaluated the framework on a marine battery use case, utilizing an LR model to predict their State-of-Health (SOH) in a decentralized manner [97]. While the proposed framework demonstrates promise in advancing privacy and decentralization, it requires additional experimentation with more complex DL models, to test its scalability and effectiveness in PdM tasks, as well as considering latency requirements, as consensus mechanisms can affect FL client training time. Moreover, the integration of FL and blockchain may increase architectural complexity and its implementation, with both technologies requiring substantial computational resources.

3.3. Hull and Structural Components

The operational safety and service-related RUL of a vessel is significantly influenced by its structural hull integrity. Based on the regulations set by the International Association of Classification Societies (IACS), hull inspections are executed during ship building and throughout its operational service phase. Specifically, during the building phase, manual inspections are performed to certify the material, and manufacturing processes have been conducted in accordance with relevant regulations and standards, while operational inspections are conducted either based on a reactive maintenance or preventive maintenance schedule on shipyards [14,98–100]. Furthermore, during a vessel's operation, failures on the hull structure manifest are due to the following:

- Corrosion: Corrosion is a natural and unavoidable phenomenon occurring at the outer surface of a ship's hull. Specifically, through the hull's prolonged interaction with the marine environment, different electrochemical reactions cause the metallic materials to oxidate, leading to the minimization of both structure density and flexibility [9,98]. An example of hull corrosion is showcased in Figure 7.
- Biofouling: Biofouling is a common problem found on the submerged surfaces of the hull, causing the formation of slime-type structures from marine organisms (e.g., algae, etc.), which has a negative impact on vessel performance efficiency (i.e., drag, increased fuel consumption) [101–103].
- Structural Fatigue: Structural deformations of ship hulls are, in the majority of cases, caused by repeated contact damage, such as that created by waves, berthing operations, or even from excessive stress from cargo overloading [98].
- Coating Deterioration: Coating layers are applied in hull surfaces, enabling their protection against biofouling and corrosion. However, the antifouling layer will gradually deteriorate due to prolonged exposure to the marine environment, leading to structural risks [98,104].

With recent developments in digitalization and ML, alternative hull inspection approaches, such as PdM, have been introduced in the literature.

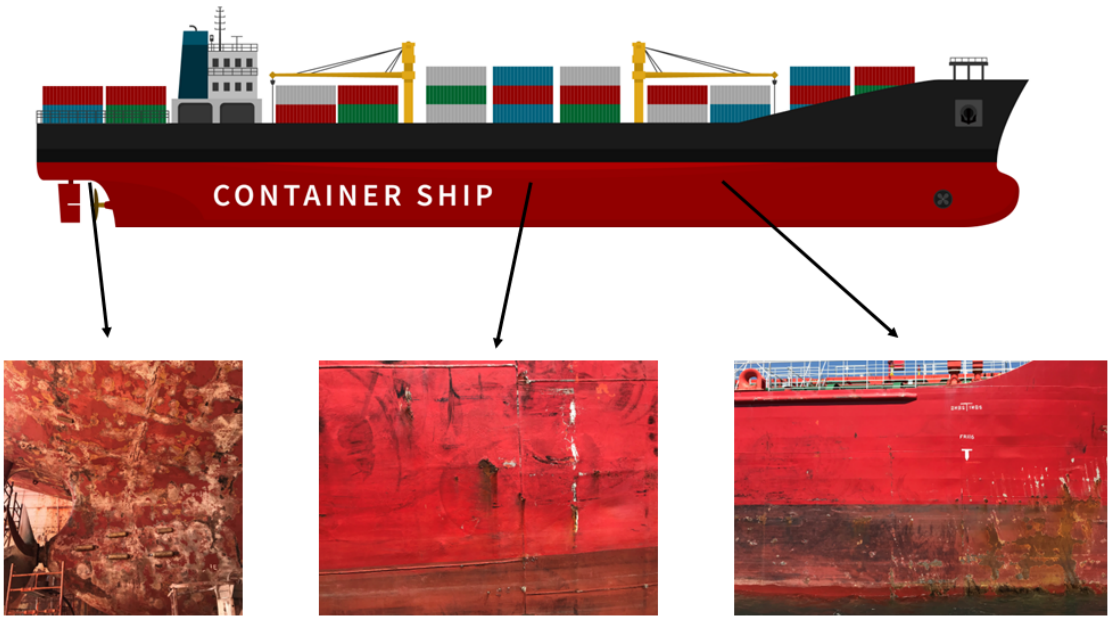

Figure 7. Illustration of marine vessel hull corrosion [105].

For instance, Juang et al. developed a stepwise regression model for predicting the corrosion rate of carbon steel pipes, taking into account the flow velocity and temperature [106]. Ji et al. proposed a comprehensive framework to predict the state of acrylic resin coatings in various marine conditions. The authors integrated key coating indicators (i.e., gloss, color, etc.) into different models (LR, genetic algorithm, and generalized Eyrin models) as an input, predicting an aging index. According to the results of the study, the genetic algorithm achieved the highest predictive performance [107].

Tree-based methods have also been utilized in evaluating hull structures, with Decision Trees (DTs) being the most widely known approach. DTs and their ensembles are non-parametric supervised learning models being used for both classification and regression related tasks, being favored due to their simplicity and interpretability [108]. Toward this end, Ji et al. introduced a hybrid framework to predict the corrosion state of hull steel structures. Their approach initially uses an LSTM model predicting dynamic marine environmental parameters over time. The predicted outputs are then combined with operational Automatic Identification System (AIS) data to generate new input vectors. These vectors are then processed by a Random Forest (RF) model optimized with PSO [109]. Another promising example of using a tree-based PdM model includes the work of Pereira et al. In this work, the authors studied the prediction of corrosion in floating production storage and offloading (FPSO) hull structures, employing a regression tree-based model, considering variables like the element material and age, the tank type, and the fluid stored [110]. While RF models avoid overfitting, handling noisy data and outliers better than a single tree and providing feature import scores, they can be less interpretable than DTs. In addition, RFs can perform exceptionally, even on small structured tabular datasets (e.g., sensor data); however, their performance may plateau as training data increases, in contrast to more complex DL configurations that can improve their prediction accuracy [111–114].

Furthermore, authors have also employed ANNs to model degradation patterns on hull structures. For example, Cipollini et al. proposed a multiclass regression model based on a Deep ANN architecture to estimate the state of various vessel components, including the hull, propeller, GT, and GT compressor of a CODLAG propulsion system [115]. Similarly, Spandonidis et al. developed an ANN model to evaluate degradation of propeller–hull systems. The approach was benchmarked against the industry standard ISO-19030 [116]

using operational data from a bulk carrier, with experimental results showcasing ANNs superior predictive performance [117]. Both of these examples demonstrate the effectiveness of ANNs in PdM applications for hull structures. However, the use of ANN-related configurations also presents several challenges, including the need for extensive labeled datasets to generalize effectively and their lack of transparency on their reasoning, being considered as "black boxes", further affecting trust among operators [118,119].

4. Current Challenges and Future Directions

Although PdM systems have the potential to improve both operational efficiency and vessel downtime, several challenges remain, hindering real-world adoption.

For instance, the absence of uniform standards and regulatory frameworks for PdM technologies affects adoption. This gap is attributed to the slow pace of adoption of advancements in the industry [9,11]. To this end, future efforts should focus on developing guidelines and standardized protocols based on open source tools to address this gap, ensuring consistency and trust in the technology. Examples include the recently established guidelines for cybersecurity and autonomous ships by the IMO and the American Bureau of Shipping [120,121].

A necessity for training accurate PdM systems is the availability of high-quality operational vessel data. Currently, publicly available maritime run-to-failure datasets are limited, enabling the oversaturation of research on specific components (e.g., GT decay prediction), while others remain underexplored. To address this, future research should explore generative AI techniques and digital twins to generate synthetic data and promote academia–industry collaborations (e.g., Data Spaces, etc.). Furthermore, prediction accuracy can also be affected due to data drift (i.e., transformation of the statistical properties of input data), resulting from upgrades, modifications, or even sensor degradation occuring onboard ships [11,122]. To address this and ensure system reliability, research should be conducted on automated data monitoring techniques and model training strategies (e.g., online learning, incremental training, FL, TL, etc.) [11,123,124].

Significant risks also arise from the use of CML-driven PdM applications. CML models introduce substantial security threats, communication inefficiencies, and challenges related to scalability and onboard computational limitations [125,126]. Future research should focus on developing PdM solutions based on decentralized ML paradigms (e.g., FL), utilizing the strengths of TL. Specifically, this could involve the exploration of different FL aggregation techniques combined with encryption methods, client selection strategies to optimize training, and the design of solutions taking into account the available computational resources onboard vessels.

Finally, the lack of transparency on model reasoning, further affects adoption in the maritime domain. In detail, ML solutions often function as "black boxes", creating a lack of confidence among operators as they are unable to interpret their reasoning [127]. To address this, regulatory frameworks such as the European AI Act have been introduced. Future research on PdM systems should focus on aligning with AI regulations, integrating model explainability methods into existing systems, and enhance transparency in their decision-making [128,129].

5. Conclusions

This review provides a detailed examination of the current PdM landscape, shaping maintenance practices in the maritime industry. Key contributions of this work include the literature analysis of PdM approaches applied to critical systems such as propulsion, auxiliary, electric, hull, and structural components. In detail, based on the analysis of this work, no single approach can serve as a one-size-fits-all solution, with each system

requiring a tailored approach, taking into consideration each model's unique advantages and limitations.

The work also identifies critical challenges that hinder adoption and discusses potential directions to overcome them. Specifically, while DL models have been documented to outperform traditional ML models in terms of scalability and predictive performance, they require large, high-quality labeled datasets to train effectively and prevent overfitting. A current limitation in the maritime PdM domain is the scarcity of publicly available run-to-failure datasets, which has resulted in research to only focus on specific components. In addition, the majority of works examined, focus on employing CML-based learning, introducing significant risks, such as data privacy concerns, security vulnerabilities, and scalability issues. As such, FL and TL are also introduced in this work as promising alternatives that could mitigate some of these challenges and drive research in a more decentralized direction. Furthermore, to ensure compliance with emerging regulations, the lack of transparency in decision-making is also discussed, with the integration of model explainability techniques into PdM systems being essential to enhance trust among operators.

In general, PdM has immense potential to increase vessel lifespans and improve safety standards in the maritime domain, presenting innovative opportunities for stakeholders to lead advancements in the field.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
AIS	Automatic Identification System
CESs	Cyber-Enabled Ships
CODLAG	Combined Diesel–Electric and Gas
DE	Diesel Engine
DL	Deep Learning
DT	Decision Tree
EGT	Exhaust Gas Temperature
ESS	Energy Storage System
EWMA	Exponentially Weighted Moving Average
FL	Federated Learning
FNN	Feed-Forward Network
FPSO	Floating Production Storage and Offloading
GT	Gas Turbine
HE	Hybrid Electric Propulsion
HFO	Heavy Fuel Oil
HVAC	Heating, Ventilation, and Air Conditioning
IACS	International Association of Classification Societies
ICE	Internal Combustion Engine
IMO	International Maritime Organization
IoS	Internet of Ships
IT	Information Technology
KF	Kalman Filter

LAR	Least Absolute Residual
LFP	Lithium–Iron–Phosphate
LNG	Liquefied Natural Gas
LNMC	Lithium–Nickel–Manganese–Cobalt–Oxide
LR	Linear Regression
LSTM	Long Short Term Memory
LTO	Lithium–Titanium–Oxide
MCN	Maritime Communication Network
MDO	Marine Diesel Oil
ML	Machine Learning
OT	Operational Technology
PCA	Principal Component Analysis
PdM	Predictive Maintenance
PF	Particle Filter
PM	Preventive Maintenance
PSO	Particle Swarm Optimization
RF	Random Forest
RL	Reinforcement Learning
RM	Reactive Maintenance
RNN	Recurrent Neural Network
RUL	Remaining Useful Life
SCC	Shore-based Control Center
SOH	State-of-Health
SOLAS	International Convention for the Safety of Life at Sea
TL	Transfer Learning
TTF	Time to Failure
TTR	Time to Repair

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